

Balanceè SmartPump

Everything the SmartPump V1 team needs — tracker, docs, and execution plan — in one place.

≡ Show menu

BALANCEÈ SMARTPUMP · ANDROID SCREEN SPEC

Every screen. *Every state. Every flow.*

Complete visual specification for Ekomobong. All 5 transaction flows, all pump display states, all attendant UI actions. Build exactly what you see here.

Flow 1 — Fixed Pre-pay Flow 2 — Fill-up Cash Flow 3 — Fill-up Digital Flow 4 — Cash Fixe

UPDATED COMPONENT SPEC

What changed. *What stays the same.*

The 6 components are unchanged physically. Two components have updated logic requirements based on the fill-up mode. Everything else is identical to the original spec.

COMPONENT A — LOGIC UPDATES (EKOMOBONG)

A

Android Display Unit — updated logic

LOGIC UPDATED

Hardware unchanged. New requirements: the Android app must support both transaction modes, the attendant swipe-up menu, nozzle shutoff detection via pulse timeout, and dynamic QR generation for post-fill-up amounts.

Hardware	7" IP65 · 600nit · Android 11+ · USB serial — UNCHANGED
New: Transaction modes	Supports fixed/pre-pay AND fill-up/post-pay
New: Attendant UI	Swipe-up menu: FILL UP AUTHORISE / AUTHORISE CASH / CASH RECEIVED
New: Shutoff detection	Pulse timeout (3s no-pulse after flow active = nozzle shut)
New: Dynamic QR	Generated post-fill-up for exact verified amount
New: USSD SMS listener	BroadcastReceiver on pump SIM — parses GTBank / Access / Zenith

Webhook payload

mode: "fixed"|"fillup" added to POST /pump/authorise

COMPONENT C — LOGIC UPDATES (OLONADE)

C

Pulse-Tap Adapter Board — updated behaviour

LOGIC UPDATED

Hardware circuit unchanged. The adapter always counts pulses and sends them over serial. The mode (fixed vs fill-up) is managed by the Android app — the adapter does not need to know which mode it is in.

Hardware	Arduino + 4N35 optocoupler + EEPROM — UNCHANGED
Behaviour	Always counts, always sends PULSE:XXXX over serial — UNCHANGED
Fixed mode	Android stops relay when count reaches litres_authorized
Fill-up mode	Android monitors for 3s no-pulse timeout → signals nozzle shut
No change needed	Olonade's firmware is correct as-is for both modes

COMPONENT E — LOGIC UPDATES (OLONADE)

E

Relay Board — two open triggers

LOGIC UPDATED

Hardware unchanged. Two conditions now trigger relay OPEN instead of one.

Default state	OPEN (pump locked) — UNCHANGED
Close trigger	Payment confirmed (any method) — UNCHANGED
Open trigger (fixed)	Pulse count reaches litres_authorized — UNCHANGED
Open trigger (fill-up)	Android detects nozzle shutoff (3s pulse timeout) → sends GPIO signal
Open trigger (emergency)	System power loss → relay defaults OPEN — UNCHANGED

COMPONENTS B, D, F — UNCHANGED

B

Payment Panel

UNCHANGED

QR + NFC reader. Now also serves post-fill-up QR. No hardware change required — the QR content changes, not the panel.

D

LiFePO4 UPS

UNCHANGED

12V 20Ah. <20ms switchover. Bridges NEPA cuts in both transaction modes. No change.

F

Cashier Tablet

UNCHANGED

Existing Balancee dashboard + pump status module. Fill-up transactions appear in the same feed as fixed transactions. No new UI needed in V1.

